10 · Did the site redesign help? (no control group) — interrupted time series (CausalPy)

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> **Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`.**

The decision. We redesigned the homepage on a known date, for **all** traffic — so there's no control group. Conversion is up since. Real lift, or just the normal upward drift? Keep the redesign or roll it back?

Interrupted time series fits the pre-redesign trend (level, slope, seasonality) and **extrapolates it forward** as the "no-redesign" counterfactual. The gap after the change is the effect — with a band that honestly widens the further we forecast. We validate with a **placebo-in-time**, track the **cumulative** impact, and check the residual **autocorrelation** that naive ITS models understate.

7-step contract.

1.1 2 · Simulate a ground truth

Daily conversion with a gentle upward trend and weekly seasonality. The redesign lands on **day 120** for everyone and adds a **true +3pp** step. The trend is the confounder (conversion was already drifting up); the step is the effect.

```
TRUE redesign effect = +3% conversion from day 120
```

t	t	day	sin7	cos7	conversion
0	0	0	0.000000	1.000000	0.097628
1	1	1	0.781831	0.623490	0.109552
2	2	2	0.974928	-0.222521	0.113692
3	3	3	0.433884	-0.900969	0.098956
4	4	4	-0.433884	-0.900969	0.100867

1.2 3 · Identify — extrapolate the pre-trend as the counterfactual

Fit the pre-period series $f(t)$ (level, slope, weekly seasonality) and extrapolate past the intervention as the no-redesign counterfactual $\widehat{Y}_t(0)$; the effect is $Y_t - \widehat{Y}_t(0)$ afterward. Being Bayesian, the counterfactual is a posterior-predictive band that **widens with horizon** — honest extrapolation.

Identification assumption (the weakest of the quasi-experimental set): no concurrent shock at the redesign date (no competing campaign, price change, or seasonality break landing at the same time). Best when the change hit everyone at once and the pre-period is long and stable.

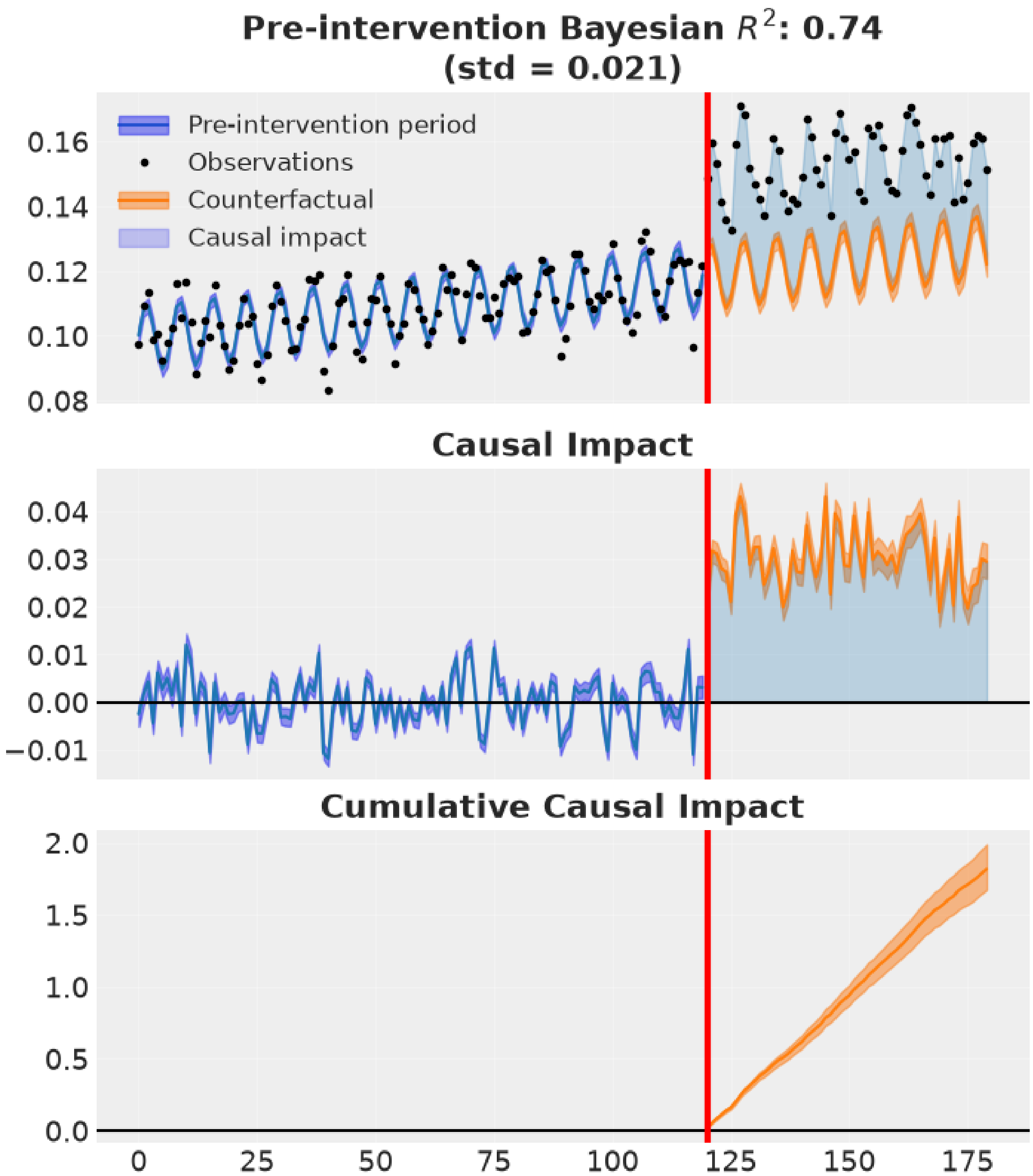
1.3 4 · Estimate — Bayesian interrupted time series

```
ITS lift +3.0% (true +3%)  ·  90% CI [+2.8%, +3.3%]
```

1.4 5 · Validate — counterfactual, cumulative impact, placebo, autocorrelation

Four panels:

1. **Counterfactual** (CausalPy's own plot) — observed vs the extrapolated no-redesign path with its widening band, and the impact below.
2. **Cumulative impact** — total extra conversions accrued since launch, with uncertainty.
3. **Placebo-in-time** — pretend the redesign happened *before* it did; the fake effect should be ~0.
4. **Residual autocorrelation** — daily series are autocorrelated; if the model's residuals still are, the uncertainty is understated.




```
recovered average step +3.0% vs true +3%
```

```
placebo-in-time +0.0% (~0, no spurious pre-effect)  ·  lag-1 residual ACF 0.42
```

1.5 6 · Decide, in euros — keep or roll back?

```
Extra conversions/day 242  ·  projected annual value €3,537,714 [90% €3,279,543,
€3,832,971]
P(redesign helped) 1.00 -> KEEP
```

1.6 7 · Caveats

- **The no-concurrent-shock assumption is the weak link.** A pricing change or competitor move at the redesign date gets attributed to the redesign. Vet the calendar around the intervention.
- **Uncertainty grows with horizon** — the band (and cumulative-impact band) widens; long-horizon claims are genuinely less certain.
- **A control group beats extrapolation.** If even a small hold-out or comparable untreated series exists, DiD (nb 08) or synthetic control (nb 07) are stronger. ITS is the last resort when *everyone* was treated.
- **Autocorrelation.** If residuals are autocorrelated (checked above), a naive model understates uncertainty; prefer trend/seasonality structure or explicit AR terms.